Evidence Review
to Inform the Programme Guidance to Protect the Nutrition of Women and Adolescent Girls in Humanitarian Settings

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Global NUTRITION CLUSTER

Acknowledgements

This publication was prepared by the Nutrition and Child Development Group at UNICEF Programme Division in New York.

Lead authors: Bridget Fenn and Tamsin Walters.

Technical reviewers and contributors were members of the Global Nutrition Cluster Maternal Nutrition Taskforce: Megan Gayford, Emily Mates, Philip James, Diane Ashley, Kirrily De Polnay, Rimu Byadyar, Casie Tesfai, Mija Ververs, Erin Boyd, Eric Anderson, Maggie Holmesheroan, Loulou Kobeissi, Adelaide Challier, Charity Zvandaziva, Natalie Sessions, Emily LaRose.

Design: Nona Reuter

UNICEF is grateful for the support received from the Bureau of Humanitarian Assistance, United States Agency for International Development, for the development of this Evidence Review.

Suggested citation: United Nations Children's Fund. *UNICEF Evidence Review for the Programme Guidance to Protect the Nutrition of Women and Adolescent Girls in Humanitarian Settings.* September 2024

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Contents

Acronyms **2**

Introduction **3**

1. Macronutrient supplementation / Balanced Energy and Protein (BEP) supplements **4**

1.1 Use of other Specialised Nutritious Foods (SNF) 7

2. Micronutrient supplementation **9**

3. Counselling and Nutrition education **11**

4. Nutrition-Sensitive Social Assistance Programmes (SAP): Agriculture asset transfers, in-kind transfers and cash/voucher transfers **16**

4.1 Agriculture asset transfers 16

4.2 Food assistance – In-kind and cash and voucher (CVA) 20

5. Women’s empowerment/ resilience **24**

5.1 Applying multiple approaches 26

Research gaps and limitations **29**

Recommendations for research priorities **30**

Endnotes **31**

Acronyms

ANC	Antenatal Care
BEP	Balanced Energy and Protein
BMI	Body Mass Index
CCT	Conditional Cash Transfers
CVA	Cash and Voucher Assistance
EHFP	Enhanced Homestead Food Production
FBF	Fortified Blended Food
FV	Food Vouchers
GAP	UN Global Action Plan
GFD	General Food Distribution
GWG	Gestational Weight Gain
Hb	Haemoglobin
IFA	Iron Folic Acid
IMAM	Integrated Management of Acute Malnutrition
IPD	Individual Participant Data
KAP	Knowledge, Attitudes, and Practices
LBW	Low Birth Weight
LMICs	Low-Middle-Income Countries
LNS	Lipid Nutrient Supplements
MDD	Minimum Dietary Diversity
MMN	Multiple Micronutrient
MMS	Multiple Micronutrient Supplements
MUAC	Mid-Upper Arm Circumference
NEC	Nutrition Education and Counselling
PBWG	Pregnant and Breastfeeding Women and Girls
PLA	Participatory Learning and Action
PNC	Postnatal Care
RCT	Randomized Control Trial
RUSF	Ready to Use Supplementary Food
SAP	Social Assistance Programme
SBC	Social and Behavior Change
SGA	Small for Gestational Age
SNF	Specialized Nutritious Foods
SQ-LNS	Small-Quantity Lipid Based Nutrient Supplements
UCT	Unconditional Cash Transfers
UNHCR	United Nations High Commissioner for Refugees
UNIMMAP	United Nations International Multiple Micronutrient Antenatal Preparation
WASH	Water, Sanitation and Hygiene
WFP	World Food Programme
WHO	World Health Organization

Introduction

This is a supplementary document to the 'Programme Guidance to Protect the Nutrition of Women and Adolescent Girls in Humanitarian Settings.' The nutrition intervention areas covered by this evidence review are:

1. Macronutrient supplementation/Balanced Energy and Protein (BEP) supplements

2. Micronutrient Supplementation

3. Nutrition Education and Counselling

4. Nutrition-sensitive social assistance programmes (SAPs)

5. Women's empowerment/resilence.

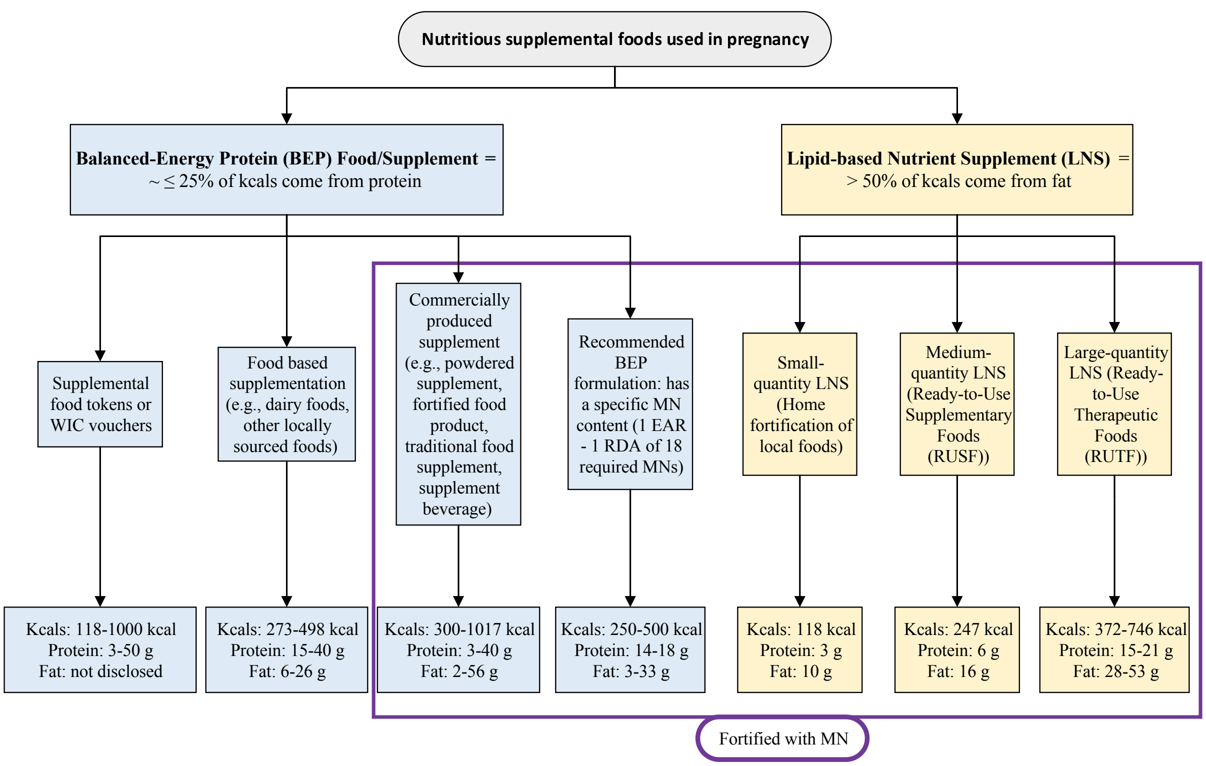
1. Macronutrient supplementation / Balanced Energy and Protein (BEP) supplements

Introduction and pathway

Products and interventions with varying compositions of macronutrients and micronutrients have been developed to improve the nutritional status of PBWG in nutritionally at-risk populations. These are broadly referred to by the collective term, specialised nutritious foods (SNFs) and are detailed in Figure 1.

Balanced energy protein (BEP) supplements and lipid-based nutrient supplements (LNS) are SNF interventions that may be given to PBWG at risk of undernutrition in food-insecure settings. BEP supplements are defined by WHO as supplements where protein provides less than 25% of total energy content.[1] BEP is a generic term for supplements that exist in various forms, including fortified cereals, biscuits, dairy products, beverages, or powdered supplements in sachets and can be made using locally sourced ingredients.

FIGURE 1. **Specialised Nutritious foods used in pregnancy as supplements**



Source: Ciulei, M.A. et al. (2023). Nutritious Supplemental Foods for Pregnant Women from Food Insecure Settings: Types, Nutritional Composition, and Relationships to Health Outcomes. Current Developments in Nutrition, 7(6).

LNS provide >50% of energy from fat (including essential fatty acids).[2] When LNS is provided in medium- and large-quantity, they can be considered a 'fortified BEP'. Both BEP and LNS products are typically (though not necessarily) fortified with micronutrients or administered along with IFA or multiple micronutrient supplements (MMS) in programming.

Nutrient content targets for food supplements for use by PBWG in food insecure contexts (LNS-PBWG) have recently been developed following a consultation led by the Bill and Melinda Gates Foundation (BMGF). The framework recommends one supplement portion of 75g LNS-PBWG should provide between 250 to 500 kcals, MMS as per the United Nations International Multiple Micronutrient Antenatal Preparation (UNIMMAP) formulation and 50% of additional protein requirements in the third trimester (14–18g) of pregnancy, although this dosage is dependent on the context and risk of malnutrition.[3]

UNHCR and WFP 2011 guidelines for selective feeding in humanitarian emergencies[4] recommend admission of PBWGs with an infant under 6 months to supplementary feeding programmes when MUAC <23cm, noting that often MUAC <21cm is used in humanitarian contexts and the chosen cut-off depends on caseload and available resources. Discharge is recommended either six months after delivery or when MUAC exceeds the admission criteria. Details on the type and content of supplementary food are limited in the guidelines.

The Lancet Maternal and Child Nutrition Series Update 2021 recommends BEP supplementation for undernourished women in pregnancy, especially in food-insecure populations.[5] BEP supplementation for pregnant women in undernourished populations has also been included in the UN Global Action Plan (GAP) framework on child wasting as one of the priority actions to reduce LBW and child wasting[a].

WHO recommends BEP supplements during pregnancy in settings with a high prevalence of underweight pregnant women[b] (20% is considered a high prevalence of underweight in women assessed by BMI).[6] WHO is currently in the process of development of BEP implementation guidance. It is anticipated that this will include guidance on optimal product composition, targeting and duration of supplementation / discharge criteria in the context of routine ANC.

What the evidence says

Summary:

- The evidence of BEP impact on maternal outcomes, such as maternal anaemia and gestational weight gain is mixed, though promising. The majority of studies on BEP have focused on infant outcomes with studies showing reduced rates of stillbirth, LBW and SGA and increased mean birth weight.
- Generating conclusive evidence from systematic reviews of available studies has been further complicated by the wide variation in supplementation modalities, including timing, duration, dosage, supplement composition and target population. The term 'BEP' refers to a range of products that differ in nutritional composition.
- It is postulated that providing BEP supplementation earlier (pre-conception or as early as possible in pregnancy) and for a longer duration may lead to better outcomes for the mother.
- There is a lack of BEP supplementation trials specifically in adolescents.

A 2023 review by Ciulei et al found that the foods or products defined as BEP ranged in calories (118–1017 kcals), protein (3–50 g), and fat (6–57 g).[7] The review summarised five recently published systematic review and meta-analysis studies (n = 20 trials), which mainly concluded that BEP supplementation was beneficial for improving *infant outcomes* (compared with no BEP). However, the authors also concluded that additional research is necessary to clarify the BEP supplement composition needed to achieve positive maternal outcomes, which pregnant women would benefit from BEP, and clarify the comparison group (only half the studies indicated that the comparison group had received iron or that iron was part of routine antenatal care). In response, the authors are conducting a prospective, individual participant data (IPD) meta-analysis which will include new, ongoing or recently completed BEP trials, which are all compared with IFA. As part of this work, BEP supplementation given during breastfeeding will also be examined.[8]

a The Global Action Plan on Child Wasting is a joint initiative of the Food and Agriculture Organization (FAO), the office of the United Nations High Commissioner for Refugees (UNHCR), the United Nations Children's Fund (UNICEF), the World Food Programme (WFP) and the World Health Organization (WHO).

b The GDG stressed that this recommendation is for populations or settings with a high prevalence of undernourished pregnant women, and not for individual pregnant women identified as being undernourished. Undernourishment is usually defined by a low BMI (i.e. being underweight). For adults, a 20–39% prevalence of underweight women is considered a high prevalence of underweight and 40% or higher is considered a very high prevalence. MUAC may also be useful to identify protein–energy malnutrition in individual pregnant women and to determine its prevalence in this population. However, the optimal cut-off points may need to be determined for individual countries based on context-specific cost– benefit analyses.

Another 2023 review (see Box 1 below, Donnelly/WFP, unpublished[9]) examined 19 studies and found that the majority (17) admitted women or girls based on the stage of pregnancy and did not enrol them based on nutritional status: the majority enrolled participants during the second trimester; just one study enrolled at preconception and two enrolled breastfeeding women only. Only two studies specifically focused on women who were underweight, using an enrolment criterion of MUAC ≥20.6cm and ≤23.0cm (Malawi) and MUAC < 21cm (Bangladesh). Only three of the 19 studies assessed the impact of BEP supplementation on maternal outcomes: an RCT in Cambodia supplementing pregnant women with CSB+ from first trimester to delivery found significant reductions in anaemia at 36–38 weeks (OR = 0.51; 95% CI: 0.34, 0.77) but no improvements in gestational weight gain (GWG)[10]; a trial in Bangladesh examining the timing of BEP supplementation with different micronutrient alternatives found that mean maternal haemoglobin level was unexpectedly lower in the early vs usual invitation groups to the government food supplementation programme[11]; and an RCT among moderately malnourished (defined as MUAC ≥20.6cm and ≤23.0 cm) pregnant women 18 years and above in Malawi found that ready to use supplementary food (RUSF) improved maternal weight gain compared with CSB+ with UNIMMAP (a multiple micronutrients tablet)[12].

A 2022 scoping review of systematic reviews and meta-analyses of the impact of 27 pregnancy interventions on maternal outcomes also found that nutritional interventions such as micronutrient supplementation and BEP improved outcomes of maternal anaemia and GWG, particularly in deficient populations[13]. While noting that BEP supplementation trials have previously found no increase in weekly GWG (mean difference: 18.63g/wk, 95% CI: –1.81, 39.07, 9 RCTs, n = 2,391), a meta-analysis that included one additional trial found a small but significant increase (mean difference: 20.74g/wk, 95% CI: 1.46–40.02, 10 RCTs, n = 2,571). No review found BEP to have any effect on the risk of pre-eclampsia.

A 2021 review of 11 RCTs conducted in South Asia[14], concluded that BEP supplementation combined with nutrition education is promising but needs to be carried out for a longer duration starting with pre-conception and at scale to have a better impact on both GWG and reduction of LBW. The type of supplement varied across the studies, which precluded conclusive recommendations.

Two systematic reviews of evidence-based nutrition interventions for adolescents found no systematic reviews of food/BEP supplementation specifically in adolescents[15,16].

BOX 1

Findings from a Review of the use of Balanced Energy Protein (BEP) supplementation for Pregnant and Breastfeeding Women and Girls (PBWG). October 2023.

A mapping of current WFP and national government BEP programming (WFP/Donnelly, 2023 (unpublished)) found that there was a broad definition of BEP supplements and a variety of products provided to PBWG. All provided 10-20% energy from protein and were fortified with multiple micronutrients. The study concurred with other published studies that the use of different formulations and different ration sizes across trials and programmes, has complicated the consolidation of evidence to inform development of guidance on BEP supplementation.

The programme mapping discovered that while countries prioritised locations for intervention based on the severity of the context, data on underweight among PBWG or birth outcomes were not used routinely to target interventions as they were usually not available. Enrolment was usually based on a MUAC<23cm, or <21cm in resource-constrained settings. Eligibility for enrolment in preventative programmes in humanitarian contexts was from pregnancy up until 6 months postpartum, with women typically enrolled in the second trimester or later. Across the 46 countries where WFP was supporting preventative BEP, all programmes used Super Cereal (CSB+/WSB+) or Super Cereal Plus (CSB++/WSB++). WFP internal guidance advises a ration size of 100-200g per person per day of fortified blended foods (FBF) based on the additional nutrient requirements of pregnancy and breastfeeding. However, key informants indicated that larger ration sizes were often given to account for sharing within the family.

The review also pointed out that blanket targeting of preventative supplementation programmes for PBWG was not always possible, in which case implementers applied additional household vulnerability or individual risk criteria. Many of the humanitarian programmes prioritised PBWG in households receiving general food assistance based on the presumption that they were the least food secure; one government programme prioritised support by adding risk criteria based on age of the mother, stage of pregnancy and anthropometric indicators. Moreover, ration sizes were sometimes adjusted based on the availability of additional support provided to the household, funding constraints or in adherence to national guidelines.

Variations were seen in prevention programming in terms of objectives, duration, eligibility criteria and delivery platform. Where prevention programming was most closely linked to general food assistance platforms, there were advantages in terms of low cost and ability to scale up rapidly, whereas programming linked to health services reported benefits in increasing the uptake of other services and ensuring a package of care was provided.

Many of the government programmes supplementing mothers using local BEP formulations had been in place for several decades, implemented through health services, but lacked strong documentation about their successes and challenges.

The WFP review proposed a list of research priorities which included:

1. More in-depth analysis/evaluation of the different government-led programmes to better understand successes and challenges of delivering BEP at scale and their capacity for shock response.
2. Agreement on a specific nutritional definition of BEP which includes the expected minimum and maximum percentage of energy from protein and micronutrients.
3. More research on different types of BEP supplement and their impact, in particular fortified blended foods (FBFs) given their large-scale use for PBWG.
4. More research into the quantities/doses given in respect of the nutrient requirements of pregnant and breastfeeding adolescents.
5. Research into the impact of supplementation on breastfeeding mothers and outcomes relevant to the recovery of mother from pregnancy and childbirth.

Research gaps

There are considerable gaps in evidence on the provision of BEP to PBWG, not least because studies frequently focus on infant outcomes, omitting outcomes on the PBWG themselves, but also due to varying intervention modalities with regard to supplement composition and quantity provided, stage of pregnancy of enrolment and baseline data on PBWG prior to supplementation.

1.1 USE OF OTHER SPECIALISED NUTRITIOUS FOODS (SNF)

Introduction and pathways

Small-quantity lipid-based nutrient supplements (SQ-LNS), which provide high-quality protein, micronutrients, and energy, have accumulated an evidence base over the past 10 years for their role in prevention of malnutrition. However, the majority of studies have focused on children. Evidence on LNS for use in pregnancy is still accumulating. There is some overlap between LNS and BEP, as mentioned above: some LNS products qualify as BEP, while 20g SQ-LNS does not.

What the evidence says

Summary:

- Evidence on the benefits of SQ-LNS for PBWG is mixed and inconclusive.
- Currently, there is no evidence to suggest benefits for pregnant women of SQ-LNS in comparison with MMS.
- There is a dearth of studies specifically focused on outcomes of LNS for PBWG.

There are three categories of LNS products: small-, medium-, and large-quantity formulations. A 2023 review reported results from 3 systematic review and meta-analysis studies (n = 5 trials) which indicated that LNS is beneficial when compared with IFA but not to MMS in outcomes including gestational duration and infant anthropometry.[17] However, the authors noted that the LNS evidence is limited, interventions have included supplements with a wide range of energy (118–746 kcals), protein (2.6–20.8g), and fat (10.0–52.6g) and there is a need for more homogenous LNS evidence to clarify whether or not the additional calories and macronutrients in LNS offer improved outcomes in comparison with MMS for undernourished pregnant women. They concluded that comparisons of BEP with MMS or LNS have been inadequately studied and deserve attention.

A 2018 Cochrane review that analysed the impact of LNS irrespective of composition, dose, or duration of supplementation, found that LNS in pregnancy had no benefit for maternal (or birth outcomes) when compared to MMS.[18] However, both IFA and MMS were better at reducing maternal anaemia when compared to LNS. The review **did not find any trials for LNS given to pregnant women in emergency settings** and the evidence came from a small number of trials, with one-large scale study (conducted in community settings in Bangladesh) driving most of the impact. Authors cautioned that effect sizes are too small to propose any concrete recommendations. This finding was reiterated in a 2022 review[19]: LNS supplements have been studied as a potential vehicle for MMS in pregnancy. In a review of four trials, LNS was not found to increase weekly GWG or reduce maternal mortality when compared to IFA or MMS alone.

A 2018 trial on dairy supplements in Guinea-Bissau found that following 12 weeks of treatment, women (a mixed group of mothers of children U5, pregnant and breastfeeding women) in villages randomised to receive a higher dairy RUSF (RUSF-33%) had an increase in MUAC compared to women receiving a lower dairy product (RUSF-15%) and to those in the control group.[20] Those on the higher dairy supplement had increased maternal weight, height, BMI, and MUAC. The intervention was also correlated with reduction in illness (diarrhoea, malaria, colds). The authors concluded that the higher dairy amount was important for mothers' nutrition. Both RUSFs had a protective effect on the mother's iron status compared to the controls, in whom haemoglobin decreased considerably.

A 2016 cluster-randomised effectiveness trial including 4011 women in Bangladesh found that LNS provided during pregnancy did not affect maternal anthropometric indicators in the overall sample but increased MUAC among women aged ≥25 years and those with lower stature and weight gain among multiparous women aged ≥25 years.[21]

Research gaps

There are significant gaps in the evidence base for understanding the potential benefits of LNS for improving nutritional outcomes of PBWG. More evidence from trials using homogenous LNS interventions is required to clarify the potential advantages of LNS over MMS and BEP in humanitarian settings with populations of undernourished pregnant women and girls. Examination of the benefits of LNS for breastfeeding mothers is another area for research.

2. Micronutrient supplementation

Introduction and pathway

The provision of micronutrient supplements aims to fill the gap between the higher micronutrient requirements of pregnancy and the low intakes frequently observed in humanitarian settings, due to reduced access to healthy, diverse and balanced diets.

A 2007 joint United Nations (WHO, WFP, UNICEF) statement advises that *"due to the increased risk of multiple micronutrient deficiencies in humanitarian emergencies, PBWG should take a daily MMS to meet their recommended nutrient intakes".* This supplement should be taken in addition to any fortified foods received.

UNICEF's programming guidance on maternal nutrition[22] states that countries may consider introducing MMS, especially in settings where women's diets are poor and anaemia, micronutrient deficiencies and/or LBW are prevalent. In settings where MMS are provided during pregnancy, women can continue taking them during the postnatal period.

The Lancet 2021 series recommended that MMS containing IFA should be provided in the antenatal period (Keats, 2021). In 2021, MMS for pregnant women were added to WHO's 22nd Essential Medicines List. A 2023 Lancet series[23] recommended provision of MMS as one of eight proven preventive interventions[a], that if fully implemented in 81 low-income and middle-income countries, could prevent 5.202 million babies being born too small and vulnerable each year.

Additional relevant WHO recommendations[24] for micronutrient supplementation in PBWG include[b]:

- Daily oral IFA supplementation as part of antenatal care, containing 30–60mg of elemental iron and 400µg of folic acid to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth. (Note: *this recommendation is not applicable if MMS is provided instead).*
- Daily calcium supplementation for pregnant women and adolescent girls in populations with low dietary intake, for the prevention of pre-eclampsia.
- In contexts where vitamin A is a severe public health problem, vitamin A supplementation is recommended for the prevention of night blindness in pregnant women and adolescent girls, to a maximum dosage of 10,000 IU per day or a weekly dose of up to 25 000 IU[c]. Vitamin A has also been associated with a reduction in anaemia during pregnancy. Vitamin A supplementation is not recommended for postpartum women or adolescent girls, as the available evidence does not support a reduction in either maternal or infant morbidity or mortality.

WHO stresses that health educators need to understand micronutrient nutrition and also regional and local variations in the diet, different cultural practices, different methods of food processing and meal preparation, and economic constraints when seeking to manage anaemia and implement micronutrient interventions.[25]

What the evidence says

Summary:

- In recent studies, MMS has been shown to provide benefits over IFA for pregnant mothers in terms of gestational weight gain, in addition to reducing adverse outcomes for newborns (stillbirths, preterm births, SGA and LBW), with greater benefits conferred among infants of women who are anaemic and underweight.

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a MMS, BEP supplementation, low-dose aspirin, progesterone provided vaginally, education for smoking cessation, malaria prevention, treatment of asymptomatic bacteriuria, and treatment of syphilis.

b In addition, WHO recommends weekly IFA supplementation (WIFAS) as a public health intervention in menstruating women and girls living in settings where the prevalence of anaemia is 20% or higher, to improve their haemoglobin concentrations and iron status and reduce their risk of anaemia. In settings where anaemia is highly prevalent (40% or higher), daily iron supplementation is recommended. WIFAS can be implemented in malaria-endemic areas but should be conducted only in conjunction with measures to prevent, diagnose and treat malaria. In an emergency setting, there are no specific recommendations for intermittent IFA supplementation. Actors are encouraged to include WIFAS as part of the Nutrition in Emergencies (NiE) interventions and while preparing the Humanitarian Response Plans (HRPs) to develop a procurement catalogue and facilitate the immediate response including logistics of supplement provision.

c A single dose of a vitamin A supplement greater than 25 000 IU is not recommended as its safety is uncertain.

- The minimum number of MMS tablets required for pregnant women/girls and optimal timing and duration of supplementation is still under debate[a].
- More experience from programming MMS interventions is required, particularly from humanitarian contexts, to better understand and advise on optimal modalities for reaching PBWG in a timely way and ensuring good coverage, uptake and adherence to supplementation protocols.

The 2023 Sight and Life MMS report,[26] reviewing the latest studies on MMS concluded that there are benefits of MMS over IFA on gestational weight gain. A 2022 IPD meta-analysis of 14 studies[27] showed that, compared to IFA, MMS resulted in a greater percentage adequacy of gestational weight gain (weighted mean difference of 0.86%; 95% CI: 0.28% to 1.44%), higher gestational weight gain at delivery (weighted mean difference of 209 g; 95% CI 139 to 280 g) and a 2.9% reduced risk of severely inadequate gestational weight gain. It did not increase the risk of excessive gestational weight gain.

Moreover, recent analyses since 2020 have demonstrated that a transition from IFA with 60mg of iron to MMS with 30mg of iron would not adversely affect maternal anaemia.

While it is generally recommended that MMS is initiated as soon as pregnancy is identified, the available evidence to date has focused on outcomes when supplementation has been initiated after the first trimester of pregnancy and less is known about the potential benefits of starting MMS before or soon after conception or continuing during breastfeeding.

High adherence to MMS has been associated with greater benefits.[28] However, the minimum number of MMS tablets that pregnant women require is still under debate. The contribution of the timing of initiation, adherence and total number tablets on the impact of MMS on birth and infant outcomes (but not specifically on women's outcomes) is being studied by an ongoing IPD meta-analysis.[29]

Poor coverage and compliance are challenges in IFA programmes for various reasons, including lack of awareness of its benefits, mistaken belief, difficulty in accessing the supplements and forgetfulness. It is, therefore, essential to build positive messaging around any programme by communicating the health benefits of supplementation to women and girls themselves as well as to influential family and community members. Mass-media campaigns designed to change social norms related to perceptions of anaemia and raise awareness of programmes, as well as sensitisation of political and community leaders can be pivotal in increasing coverage and uptake. The use of social media has potential as a platform to provide nutrition education and to motivate adolescents, increasing their knowledge and awareness and influencing attitudes.

Research gaps

The minimum number of MMS tablets required (in terms of duration of supplementation) during pregnancy is still under debate. Although MMS programming modalities are being examined for effects on birth and infant outcomes,[30] studies also need to consider outcomes on the health and well-being of pregnant women and girls.

The 2023 MMS Sight and Life report calls for better documentation and dissemination of actual experiences of MMS supplementation in humanitarian settings[31]; and a recent review to understand how MMS is delivered in humanitarian contexts found that it is usually delivered through health services, but no examples were found of stand-alone interventions.[32] It is still predominantly delivered in small-scale programmes and pilots and the MMS themselves are mostly imported, adding to concerns over its cost in comparison with IFA. Furthermore, the lack of local/context-specific evidence on its effectiveness and implementation was reported as a barrier to any switch from IFA to MMS. Coverage and adherence data was lacking for both IFA and MMS, but where it existed there were huge gaps in both coverage and uptake of MMS. The study concluded that MMS ultimately needs to be integrated into national programming if it is to be used sustainably in humanitarian contexts.

a In principle, MMS should be provided daily throughout pregnancy.

3. Counselling and Nutrition education

Introduction and pathway

WHO has recommended that counselling on healthy eating should be integral to antenatal care.[33] One of UNICEF's programmatic priorities is to improve the coverage and quality of nutrition counselling before and during pregnancy and while breastfeeding.[34]

Counselling for maternal nutrition is an interactive process between a service provider and a woman and her family during which information is exchanged and support is provided so that the woman and her family can make decisions and take action to improve her nutrition.

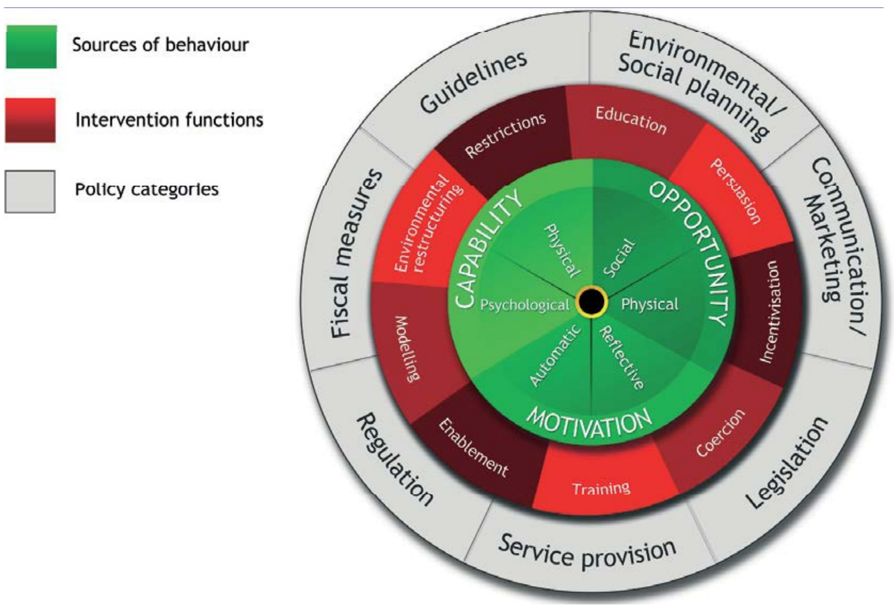
Key aspects relevant to humanitarian contexts from UNICEF's Technical Brief[35] include formulating targets, plans and budgets to expand the coverage, quality and equity of maternal nutrition counselling services for women and girls before and during pregnancy and while breastfeeding.

However, limited time, infrastructure, staff capacity and motivation often hinder or prevent the delivery of quality antenatal care in low-resource settings—and often, nutrition counselling does not happen or is not well conducted.[36]

There are various applicable models for counselling and behaviour change and a degree of distinction between interventions which aim to provide nutrition education and counselling (NEC) targeted primarily to individuals or groups of PBWG, compared with social and behaviour change (SBC) interventions which are broader in scope and aim for societal change.

Designing behaviour change interventions requires engagement with the target population, understanding their motivation to change, and adapting interventions to the contexts that facilitate change, including environment and social networks. One model used by Health Psychologists is the Behaviour Change Wheel (Figure 2),[37,38] which outlines nine behavioural intervention functions, that aims to address deficits in one or more of the three underlying human factors–Capability (physical and psychological), Opportunity (physical and social), and Motivation (automatic and reflective) that influence behaviour.

FIGURE 2. **Behaviour Change Wheel**



Source: Michie, S., van Stralen, M.M. & West, R. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. *Implementation Sci 6,* 42 (2011).

What the evidence says

Summary

- There is some evidence that NEC can improve adherence to micronutrient supplements and dietary intake during pregnancy.
- There is limited evidence on NEC or SBC interventions on outcomes of women's or girls' nutritional status; partly due to lack of rigorous research.
- There is evidence that NEC in combination with other nutrition interventions can improve outcomes.
- There is growing evidence on the need to include husbands and other family members for effective NEC; and to additionally include community members in action to expand to SBC initiatives.
- Increasing evidence is showing that pregnant and breastfeeding adolescents are a distinct group and formative research / joined-up and different approaches are needed to engage them in effective NEC programming.
- Well-designed research that is grounded in appropriate theories of behaviour change is needed to improve confidence in the effectiveness of NEC and /or SBC interventions to improve nutrition of PBWG.
- Evidence on effective NEC or SBC approaches in humanitarian settings is limited.

A 2023 systematic review[39] to collate evidence on whether nutrition behaviour change interventions are effective in improving maternal and child nutrition in sub-Saharan Africa and which functions of behaviour change interventions are associated with improvements in nutrition outcomes examined 79 articles. Many that applied behaviour change theory, communication or counselling resulted in significant improvements in household dietary intake and maternal psychosocial measures (among others). Intakes were largely not measured in the studies, so evidence could not be generated on whether the interventions improved maternal nutrient intake. However, reviewed interventions identified improvements in maternal psychosocial outcomes in relation to nutrition, including knowledge, practice, attitude, intention, confidence, capability and wellbeing.

The authors recommend incorporating behaviour change functions in nutrition interventions, specifically drawing on the Behaviour Change Wheel, through collaborations between behaviour change and nutrition experts, policy makers and programme implementers to fund and roll-out multicomponent behaviour change to improve both maternal nutritional and psychosocial outcomes. The review findings indicated that interventions comprising all three Behaviour Change Wheel intervention functions (incentives, persuasion and environmental restructuring) were most likely to be effective in improving nutritional and psychosocial outcomes.

Nutrition counselling has been shown to improve diet, adherence to micronutrient supplements, and food security in pregnant women.[40] However, studies suggest that tailored, culturally resonant NEC on diet during pregnancy and breastfeeding and weight gain during pregnancy, as well as monitoring of progress in maternal nutrition, are areas requiring attention[41]. Formative research is critical to the design and implementation of culturally resonating interventions, approaches, and messages. In addition, strengthening communication and counselling skills is necessary to address misperceptions and cultural beliefs around maternal nutrition.

A 2012 systematic review and meta-analysis of outcomes of NEC provided during pregnancy reported that significant differences between GWG in intervention and controls were only observed when NEC was provided with nutrition support (0.15kg [95% CI 0.00, 0.29]).[42] NEC substantially and significantly reduced risk of anaemia in LMICs (0.69 [95% CI 0.56, 0.85]). When stratified on category of NEC, effects were substantial and significant when NEC was provided with nutritional supplementation, usually micronutrients (0.58 [95% CI 0.44, 0.76]). Effects were less strong and only borderline significant when NEC was provided alone (0.84 [95% CI 0.70, 1.00]). There were insufficient numbers of studies providing complete data to allow authors to conduct meta-analyses on any of the maternal morbidity outcomes. Seven studies examined maternal morbidity during pregnancy including pre-eclampsia, eclampsia, PIH, bleeding during late pregnancy, haemorrhage during labour and gestational diabetes. None of the seven studies reported a significant effect of the intervention on any maternal morbidity outcome although most studies were not adequately powered to assess morbidity outcomes. The overall quality of the body of evidence was deemed low for all outcomes due to high heterogeneity, poor study designs and other biases. Studies in which NEC was part of a package of other health messages consistently reported null effects for the outcomes of interest. The authors suggested that it was likely that the NEC component was diluted or de-emphasised as other messages were added.

A mixed-methods systematic review of behavioural interventions in LMICs aimed at increasing family support for the nutrition of PBWG, infants and young children[43] included 35 peer-reviewed articles on 25 studies. Of 16 studies designed to compare quantitative outcomes, only one focused on maternal nutrition.[44] This RCT conducted in Bangladesh focused on NEC of husbands and their subsequent behavioural determinants and support to their wives in improving maternal nutrition, including adherence to IFA and calcium supplementation. In terms of improvements in the intervention group in IFA consumption, 48% of the total programme difference was explained by improved fathers' behavioural determinants and supportive activities. The attributable differences were 44% for calcium tablet adherence and 22% for dietary diversity. The authors affirmed that the important role of family support in enhancing mothers' capacity to adopt recommended practices was strengthened by qualitative evidence and by the few studies analysing relationships between support variables and nutrition practices.[45, 46]

A 2020 cluster-randomized controlled trial examined the effect of guided counselling on nutritional status of pregnant women in West Gojjam zone, Ethiopia.[47] The intervention started before 16 weeks' gestation and pregnant women in the intervention group attended 4 counselling sessions, provided in their homes using a counselling guide. Leaflets with appropriate pictures and core messages were given to women. Women in the control group received routine nutrition education given by the healthcare system. Those in the intervention group were counselled to increase portion size and frequency of meals with increasing gestational age. Counselling was also given on improving the diversity of meals, taking IFA supplements and using health services. Attitude, subjective norms, self-efficacy, perceived control, intention, knowledge and dietary practice were assessed during each counselling session, after which counselling was given based on the identified gaps and household income. Individual nutrition counselling was provided through a monthly home visit on non-working days (religious holidays and weekends), in sessions of 40 to 60 minutes. Counsellors used a client-centred approach to identify women's dietary practices and their specific needs in terms of nutrition, considering women's needs, household income and identified gaps and allowed the women to choose recommendations that were locally available, acceptable and affordable.

Women in the intervention group showed significant improvement in nutritional status at the end of the trial compared with the control group (β= 0.615, $p = < 0.001$). The mean MUAC in the intervention arm increased by 33% from the baseline while the prevalence of undernutrition was 16.7% lower in the intervention group compared with the control arm (30.6% Vs 47.3%, $P = < 0.001$). T-test results showed that the intervention improved the mean MUAC by 61%. The authors concluded that guided counselling using the health belief model and the theory of planned behaviour was effective in improving the nutritional status of pregnant women.

A quantitative randomised controlled evaluation of a Participatory Learning and Action (PLA)[a] women's group intervention, with and without unconditional food and cash transfers, examined effects on women's agency within the household in rural Nepal.[48] This study was nested within a four-arm cluster-randomised trial comparing: 1) women's groups practising PLA; 2) PLA women's groups combined with unconditional cash transfers; and 3) PLA women's groups combined with unconditional food transfers; against 4) a control arm. The authors found little evidence for an impact of PLA alone or combined with unconditional food or cash transfers on women's agency in the household. Birthweight as the primary outcome was significantly higher in the food plus PLA arm, but not in the PLA only arm or cash+ PLA arm. Weight during pregnancy was higher than the control group in all other groups, but not significantly so and no significant differences were seen in maternal anthropometry (BMI, underweight or MUAC).[49]

An extensive literature review of peer-reviewed and grey literature on maternal diet during pregnancy and breastfeeding and weight gain during pregnancy[50] found that food intake during pregnancy and breastfeeding was largely driven by personal preferences and cravings, food avoidance, food taboos, and cultural beliefs surrounding pregnancy physiology. Most mothers followed advice from trusted family members. Food consumption during pregnancy was often reduced and increasing the quantity of food during pregnancy was challenging for most mothers. However, there was considerable variation in food consumption patterns of breastfeeding mothers, from reducing food intake to eating larger amounts. Despite the opportunity to counsel mothers on diet during pregnancy at routine health facility visits (i.e., ANC) and/or visits by community health workers, only seven studies cited that nutrition information was given during these visits. The authors concluded that advice from influential family members, economic constraints, and intra-household food allocation is a barrier to achieving optimal maternal diet. This review revealed that limited evidence is available on the type and quality of information and counselling received on maternal nutrition and weight gain during pregnancy at health facility and/or community levels. The review concluded that engaging grandmothers, women elders, and other key influencers of nutrition practices, such as fathers, presents an opportunity for behaviour change at the community and household level and can encourage early and frequent ANC visits. Evidence of interventions to address a healthy maternal diet during pregnancy and the post-partum period and optimal weight gain during pregnancy is scant.

a This is a form of behaviour communication change that involves active participation of community group members, typically in four phases:
Phase 1: Identification and prioritisation of the problems facing the group members.
Phase 2: Planning solutions that are feasible and sustainable, addressing the priority problems.
Phase 3: Implementing solutions.
Phase 4: Assessing how the activities have gone and translating the lessons learnt into future planning.

A systematic review of mobile Health (mHealth) interventions targeting pregnancy intakes in low and lower-middle income countries[51] found 3 RCTs and one pre-post experimental study from India, Indonesia and Kenya. All studies examined use of supplements in pregnancy. The authors concluded that mHealth can improve intake of supplements and improve pregnant women's nutrition status; however, further evidence is required on the influence of mHealth on dietary intake.

Literature specific to adolescents

A 2020 global systematic review of health education for improving knowledge, attitudes and practices (KAP) of adolescents on malnutrition concurs on the need to support the inclusion of parents and adolescents in interventions aiming to increase KAP.[52] This review found limited evidence of a relationship between intervention duration or frequency of contact and effectiveness. However, the level of personalisation and the reach of intervention (intervention settings) showed an association with overall effectiveness of intervention. A limitation of this study was that the majority of studies were from HICs, alongside studies from Bangladesh, India and Palestine.

A systematic review in 2019 did not find any study that assessed the impact of nutrition education and counselling among adolescents (whether pregnant or non-pregnant).[53]

WFP conducted formative research with adolescents in Kenya, Cambodia, Guatemala. The findings of these studies highlight important considerations for interventions attempting to reach pregnant (and non-pregnant) adolescents with nutrition counselling and SBC. In Kenya,[54] the study found that:

- Technology platforms are a promising way to engage adolescents. However, the penetration and use of technology is highly context-specific, and differs according to social groups, age and gender. Mobile technology was a popular means of communication, although adolescents in Samburu had markedly lower access to and usage of all technology platforms.
- Engaging adolescents when they are younger (e.g., 10-14 years) is important. Normalising health facility visits for this age group can reduce stigma related to attendance and would help move away from the negative association between health facility attendance and sexual reproductive health issues.
- Ingrained gender norms related to food allocation within the household prevent girls' healthy nutrition. Raising awareness about the importance of an adolescent girl's nutrition should focus on her strength and role in the (household) economy (in terms of immediate value) and on the importance of her health for the next generation (future value).
- Engaging with key male and adult influencers is critical.
- Raising awareness around good nutrition during pregnancy needs to be discussed in adolescent forums. In parallel, initiatives should improve ANC, delivery practices and postnatal care to assuage fears around having large babies (and therefore restricting diet during pregnancy).
- Cheap, safe and healthy snack foods should be made available for pregnant adolescents, and consideration given to snacks in terms of their value as food and micronutrient supplements.

Highlights from WFP's formative research in Cambodia[55]

Multiple food taboos were identified by participants from the indigenous ethnic minorities in Ratanak Kiri, many of which related specifically to pregnant and breastfeeding women. These taboos were often enforced by strong social norms, and non-adherence risked consequences not only for the mother and her child but for the whole community. Food restrictions were reinforced by the desire to bear smaller babies to avoid a difficult labour.

Some of the most-practised food taboos in Ratanak Kiri restricted the consumption of foods that could provide valuable sources of protein and vitamins (e.g., eggs, fish, fruit, vegetables). It was also considered to be more important for first-time mothers to strictly observe food taboos than it was for women who had had multiple children. The authors suggested that this may have contributed to the difficulties many mothers reported in relation to their first pregnancy and delivery, particularly in the case of adolescent pregnancy.

To strengthen the evidence base, the authors call for disaggregating available data for adolescents and systematising routine collection of adolescent-specific data. To complement and supplement routine quantitative data, high-quality qualitative data should be collected to better understand the lived realities of adolescents, the complex root or underlying causes of their nutrition practices, and potential solutions to improve their food-related behaviours.

The study also found that the definition of adolescence at the national level was not consistent with definitions used at the community level. They concluded that interventions need to be sensitive to variables including age, gender, socio-economic status, life experiences / stages, livelihoods and ethnicity. Regardless of the terminology used, effective engagement should target groups as defined and understood at the community level.

A 2015 case study[56] on adolescent inclusion in the care group approach in Northwest state of Sokoto, Nigeria used key informant interviews and focus group discussions to explore experiences and highlight barriers, facilitators, and lessons learned. Findings included: the decision to attend the Care Group programme was made by husbands of the adolescent girls, with two of them receiving joint permission from their mothers-in-law. Lead Mothers educated the girls on appropriate foods for consumption during pregnancy and breastfeeding. The Lead Mothers also had an impact on husbands and the types of food they purchased for their households. Community Leaders reported that as a result of attending the programme, adolescent girls were accessing health services, such as ANC and PNC, more than they had previously. Husbands were influenced by the programme, along with other family members, friends and neighbours who gained knowledge from the Lead Mothers during house visits. House visits also provided Lead Mothers with opportunities to access unmarried adolescent girls who were present in the household, with nutrition, WASH, and hygiene information. While adolescents preferred adolescent-only groups, as they could freely ask questions and discuss their concerns, Lead Mothers thought groups mixed with older women would allow adolescents to learn from experienced women. The study concluded that community and family buy-in is critical when targeting adolescent girls for inclusion in Care Groups.

Maternal mental health and psychosocial support

People living in contexts experiencing conflict or fragility are at increased risk of psychological distress and mental health disorders. Studies across various countries have estimated that between 8.3 to 39% of pregnant adolescent girls experience depression, which is disproportionate to the prevalence in adult women, among whom it is 11–18%.[57, 58] There is also evidence that poor mental health among women and adolescent girls, during pregnancy and postpartum leads to inferior short and long-term health outcomes for both mother and baby.[59, 60]

No studies were found that explicitly linked mental health or psychosocial support with maternal nutrition outcomes. A 2020 review that considered effectiveness and delivery of mental health interventions in conflict settings found that most of the literature is focused on interventions targeted at children: only 26 out of 157 studies documented interventions targeted at women.[61] Of these, psychosocial support was the most frequently reported intervention, followed by training interventions and then screening (for referral/with intention to treat). The review found that psychosocial support for mothers tends to be combined with psychosocial stimulation for children or breastfeeding counselling interventions. The Inter-Agency Standing Committee (IASC) 2007 guidelines recommend that support for caregivers in humanitarian contexts involves the provision of safe spaces for caregivers to meet, support each other and discuss strategies for optimal childcare and other concerns.[62] Referral options for additional support for carers with signs of depression or severe mental health problems should be provided.

Research gaps

The overall quality of the body of evidence was deemed low for all outcomes due to high heterogeneity, poor study designs and other biases. Additional well-designed research that is grounded in appropriate theories of behaviour change is needed to improve confidence in the effectiveness of NEC and /or SBC interventions to improve women's nutrition. Cost-effectiveness research is proposed, to clarify the added benefit and sustainability of providing NEC/SBC with nutritional support and/or safety nets, especially in areas where food insecurity and gender bias may limit women's capacity to adhere to NEC/SBC messages.

Evidence from the published literature as well as from formative and programmatic research points towards the importance of the inclusion of family members in interventions, with attention to building family support in ways that fit the cultural context. More rigorously designed and implemented research would strengthen interpretation and inform development of effective, sustainable interventions. A notable gap was a lack of research on interventions focused on maternal nutrition outcomes, such as anthropometrics or anaemia.

To complement and supplement routine quantitative data, high-quality qualitative data should be collected to better understand the lived realities of PBWG, the complex roots or underlying causes of their nutrition practices, and potential solutions to improve their health, wellbeing and food-related behaviours.

4. Nutrition-Sensitive Social Assistance Programmes (SAP):
Agriculture asset transfers, in-kind transfers and cash/voucher transfers

PBWG are especially vulnerable to the negative impacts of food insecurity,[63] due to their increased nutritional requirements during pregnancy,[64] and for breastfeeding.[65] Food insecurity can be a risk factor for depression, anxiety and stress and poor-quality diets during pregnancy, as well as for breastfeeding interruption, and increases the physical and psychosocial vulnerability of pregnant and postpartum women and girls during these periods.[66] Addressing the food insecurity of the household is important to ensure the nutritional requirements of women are met.

There has been an increase in the use of nutrition-sensitive social assistance programmes (SAPs) to address both food security and nutrition outcomes. The programmes identified for potentially having an impact on PBWG and/or maternal nutrition outcomes are agriculture asset transfers, in-kind transfers and cash and voucher assistance (CVA). The evidence identified for this review was restricted to any social assistance programmes without always specifically identifying whether women were pregnant or not to allow a broader sweep of the evidence. However, much of the evidence available for agricultural assets, relates to the mother of an index child/ren under study, and doesn't distinguish whether mothers are also pregnant or breastfeeding. For CVA there was more distinction between maternal and PBWG. In any case, there was no evidence specifically for women who were breastfeeding.

Summary

- There is clear evidence of positive impacts of nutrition-sensitive SAPs on maternal diet-related outcomes; there is less evidence here for PBWG, especially those breastfeeding.
- There is limited evidence of nutrition-sensitive SAPs on maternal/PBWG anthropometric status.
- The evidence on how nutrition-sensitive SAPs may lead to improvements in maternal/PBWG nutrition outcomes is mixed and studies within the reviews largely heterogeneous.
- The pathways through which maternal nutrition is affected by agricultural-asset transfers is unclear and needs to be better unpacked in order to understand the important entry points for interventions.
- Nutrition-sensitive agricultural interventions are more likely to improve maternal/PBWG nutrition outcomes when they include strong nutrition SBC and women's empowerment components.
- Household food assistance programmes, including both household and individual transfers, are more likely to improve maternal or PBWG's nutrition outcomes if they specifically target those women, and include fortified foods/supplements and/or SBC.
- There is an overall lack of good quality evidence on nutrition-sensitive SAPs that specifically looks at maternal or PBWG nutrition outcomes in this domain, especially in humanitarian settings.

4.1 AGRICULTURE ASSET TRANSFERS

Introduction & impact pathway

"Women's work in agriculture is significant; and for some populations, their total contribution to productive activities is often greater than what more common narratives may suggest".[67]

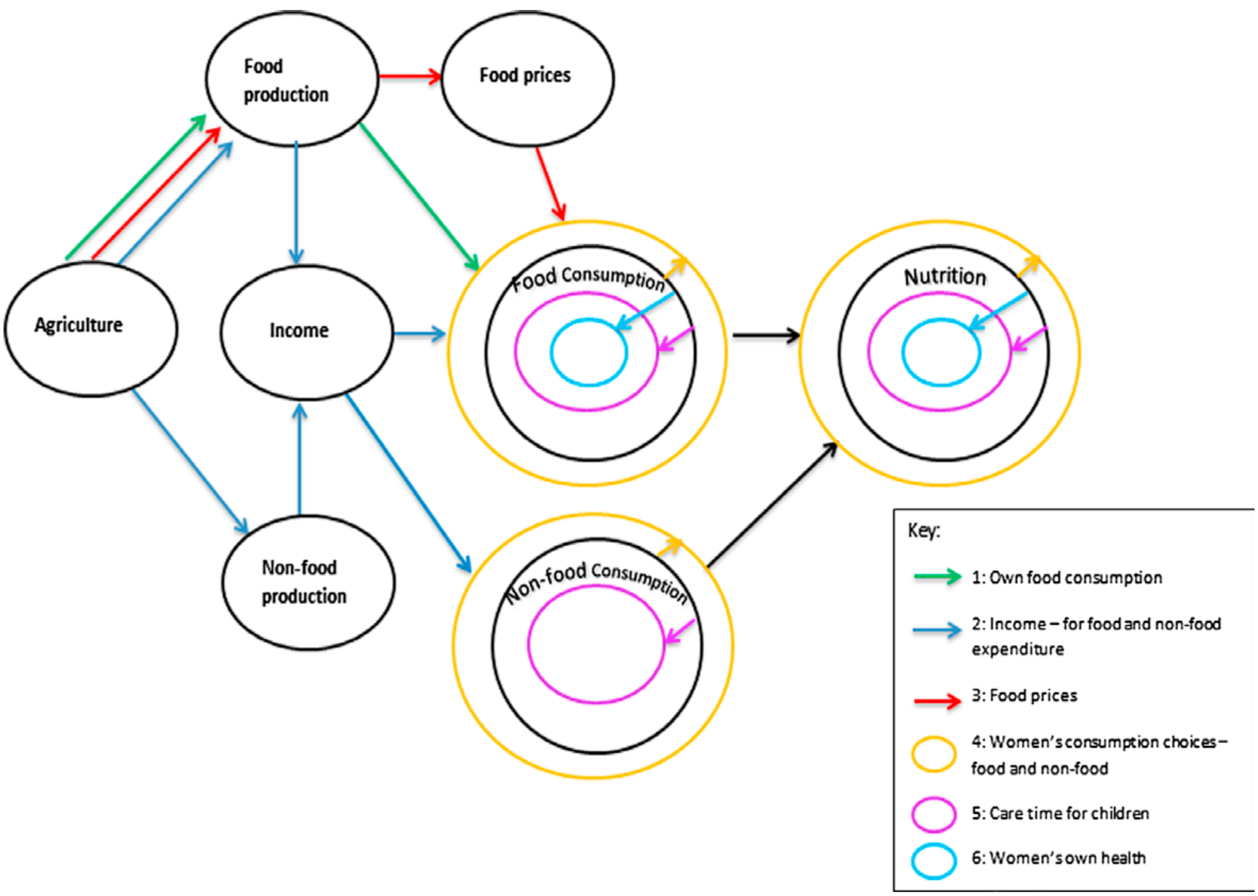
Programmes aimed at increasing household food intakes, with the aim to improve maternal nutrition outcomes, such as home food production e.g., kitchen gardens, are usually targeted at crisis-affected and vulnerable households and individuals to address food needs. However, in isolation these programmes may not be sufficient to improve maternal nutrition outcomes either because the cause of maternal malnutrition is not directly linked to food insecurity and/or because programmes do not tackle food insecurity sufficiently e.g., by not adequately taking into account the causes of food insecurity, nor intrahousehold dynamics and sharing, inequalities in decision-making, lack of cooking facilities, lack of knowledge on food storage and safety, competing non-food expenses e.g., on health.

A number of pathways have been proposed through which women's work in agriculture may contribute to improved maternal nutrition outcomes.[68, 69, 70] The framework proposed by Rao (2019) posits 6 pathways to improve nutrition status through agriculture (Figure 3); three through agricultural production: increasing the availability of nutritious foods (own consumption – Pathway 1), increasing incomes (Pathway 2), reduced prices of nutritious foods (Pathway 3), and three through women's work in agriculture: women's food and non-food consumption choices (Pathway 4), caring practices and capacities (Pathway 5) and women's own health (Pathway 6).[71,72] The authors hypothesise that women's work in agriculture may lead to improvements in maternal nutrition through Pathway 4 or deterioration through Pathways 5 and 6.

Similar pathways have been observed through enhanced homestead food production (EHFP) programmes that consist of e.g., home gardens, poultry raising, and nutrition education.[73]

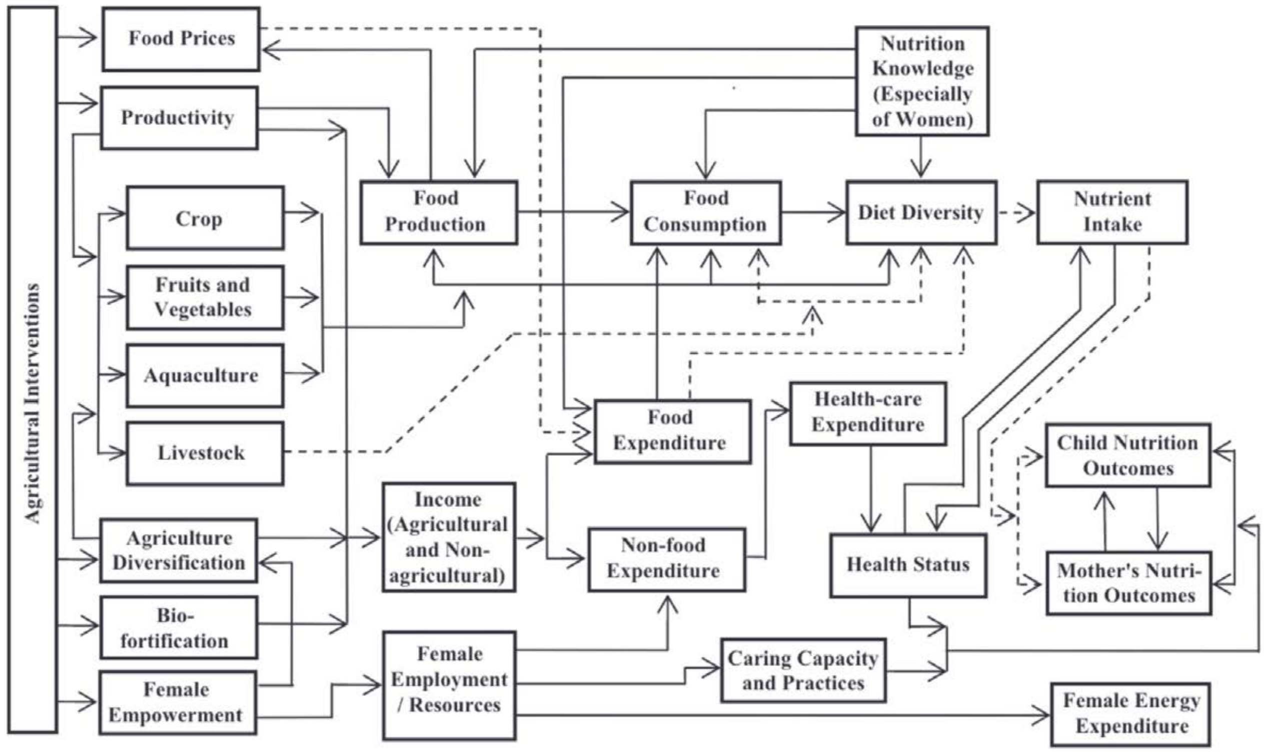
Pandey et al concludes from their review that multisectoral approaches to agricultural interventions that are also targeted at empowering women can enhance maternal nutritional outcomes (Figure 4).[74] As well, there have been some studies, all be with small sample sizes, on women's empowerment and dietary diversity that have shown that improved dietary diversity is positively associated with women's empowerment in a number of different settings.[75, 76, 77]

FIGURE 3. **Pathways to improve nutrition status through agriculture**



Source: Rao, N. et al. (2019). Women's agricultural work and nutrition in South Asia: From pathways to a cross-disciplinary, grounded analytical framework. Food Policy, 82, pp. 50–62 based on Kadiyala, S. *et al*. (2014). Agriculture and nutrition in India: mapping evidence to pathways. *Annals of the New York Academy of Sciences,* 1331, pp. 43–56.

FIGURE 4. **Pathways between agriculture intervention and nutrition**



Source: Pandey, V.L. et al. (2016). Impact of agricultural interventions on the nutritional status in South Asia: A review. Food Policy, 62, 28–40.

Figure 4 shows the pathways between agricultural intervention and nutrition.[78]

Whilst it is becoming increasingly recognised that women's empowerment is considered a crucial pathway to improved nutrition outcomes most research carried out [on women's empowerment] relates to the nutrition outcomes of children rather than the mother herself, even though the available evidence surrounding women's empowerment is more strongly associated with maternal rather than child outcomes.[79] As well the domains of women's empowerment that are associated with maternal nutrition outcomes are not the same for children's nutrition outcomes.[80]

However, it still remains uncertain through which of these pathways nutrition outcomes are mainly achieved as few studies have investigated this. The pathways linking women's work in agriculture to maternal nutrition outcomes are complex and interconnected and considering the wide heterogeneity in household dynamics and the constraints faced suggests a 'one size fits all' approach would be inappropriate.

At the same time, there is a lack of good quality and robust evidence, partly due to methodological challenges, especially from humanitarian settings, and with a larger focus on children's nutrition outcomes. Most of the evidence comes from development settings and the assumption that applying this evidence to the needs and practices of nutritionally vulnerable populations will follow the same theories and impact pathways as in other non-affected settings, needs to be further examined.[81]

What the evidence says

Five recent reviews were identified[82,83,84,85,86] although the number of individual studies identified within these reviews relative to maternal nutrition were not as many as for children's outcomes. Only the Baliki review was relevant to humanitarian settings, although with less focus on PBWG's nutrition outcomes. Two studies not fully included in the reviews are also included here as they add to the evidence,[87,88] Reviews and studies were identified on their inclusion of PBWG's nutrition outcomes and included those that were designed to increase household food production,[89,90] including homestead gardening,[91] enhanced homestead food production (EHFP) strategies[92,93] and those that observed the impacts of the contribution of women's work in agriculture on her own nutrition.[94,95]

Due to the heterogeneous nature of the individual studies pooling of data was not feasible for any of the reviews.

Dietary intakes and diversity

There is some consistent evidence of a positive association between agricultural interventions and maternal nutritional outcomes with regards to the intake of nutritious foods, especially where programmes were aimed toward the increased production of nutrition-rich crops, agricultural diversification strategies (especially vitamin A rich fruit and vegetables, including indigenous vegetables and leafy greens, and aquaculture) and those that engaged women directly e.g., homestead gardens[96,97,98,99]. For example, there is consistent and significant evidence of improved diet patterns and vitamin A intakes in mothers in programmes designed to improve the quantity and/or quality of household food production.[100] For the EHFP programmes, dietary intakes of zinc and vitamin A improved (EHFP group compared to a control group)[101] and higher intakes of iron, vitamin A, riboflavin and lower prevalence of inadequacy of those nutrients were noted when EHFP was combined with aquaculture, compared to a control.[102] Rao et al highlight the importance of a women's empowerment strategy built into agricultural interventions to have a positive impact on maternal nutrition outcomes.[103] The two individual studies both reported improvements in maternal anaemia either through EHFP+SBC[104] or technological innovation to improve crop production.[105]

The evidence on the impacts on dietary diversity is mixed. There is some evidence highlighting that homestead gardens may improve dietary diversity and increase consumption of nutrient-rich foods, but with little conclusive evidence of an impact of overall household food security,[106] although this evidence was not disaggregated by individuals. At the same time nutrition-sensitive agricultural interventions aimed at improving dietary diversity through increasing the diversity of crop production did not have an impact.[107] Evidence suggests that combining EHFP with an SBC component is crucial to improvements in dietary diversity,[108] and that homestead gardens themselves can lead to positive changes in food choices (as well as maternal nutrition outcomes and women's empowerment).[109]

Anthropometry

There was less evidence of an impact on maternal anthropometric indicators, either BMI and/or MUAC (and with fewer studies measuring MUAC). The evidence available was mixed depending on the intervention and the methods. For example, interventions that focused on the production of nutritious foods (both micro and macro-nutrients) potentially improved BMI[110,111] as did homestead gardens, and diversification of the agricultural production system towards fruits and vegetables and aquaculture.[112] Girard et al also noted the importance of multisectoral strategies to improve maternal nutritional status.[113] Whilst Olney et al included MUAC and BMI in their inclusion criteria no studies were identified that measured these indicators.[114]

Women's empowerment

There is limited, but positive evidence of the impact of women's empowerment and maternal nutrition, although this is dependent on the domain of women's empowerment under study. For example, Rao et al highlights a positive relationship between women's decision-making related to agriculture and household activities and nutrition through pro-nutrition food choices especially.[115] Another review highlighted women's autonomy in agricultural production decision-making having positive effects on maternal nutritional outcomes, but not including BMI. But that when women had control over household incomes or participated in women's groups or had improved empowerment overall there was an improvement in maternal BMIs.[116] However, there remains some concern that women's empowerment may have a negative impact on maternal BMI whereby agricultural strategies leading to greater demands on time through increased work intensity may affect a woman's health if she is not adequately compensated.[117]

Rao et al (2019) point out the lack of evidence of changes in agricultural practice on empowerment.[118] Whilst other reviews suggest that there is evidence that homestead gardens have a positive impact on women's empowerment.[119,120]

Gender-sensitive SBC, to improve mothers' nutrition knowledge, combined with nutrition-sensitive SAPS, is crucial for nutritional outcomes through all the pathways.[121]

Research gaps

There are significant gaps in the evidence between existing research and the needs and current practices in humanitarian settings for maternal nutrition.

The lack of evidence on maternal nutrition outcomes is a potent call for additional research that is robust and of good quality as well as being adequately powered to detect changes in nutrition outcomes. In which case sample sizes need to be large enough to detect changes in maternal nutritional status rather than just taking the sample of mothers of children included in the survey.

Evidence needs to be sourced from different and wider settings and take a more contextualised approach. There is less research in humanitarian settings and applying theories of change from development settings may not hold, especially in conflict or crisis-affected settings.

The pathways through which maternal nutrition is affected by agricultural-asset transfers is unclear and need to be better unpacked in order to understand the important entry points for interventions.

4.2 FOOD ASSISTANCE – IN-KIND AND CASH AND VOUCHER (CVA)

Introduction and impact pathway:

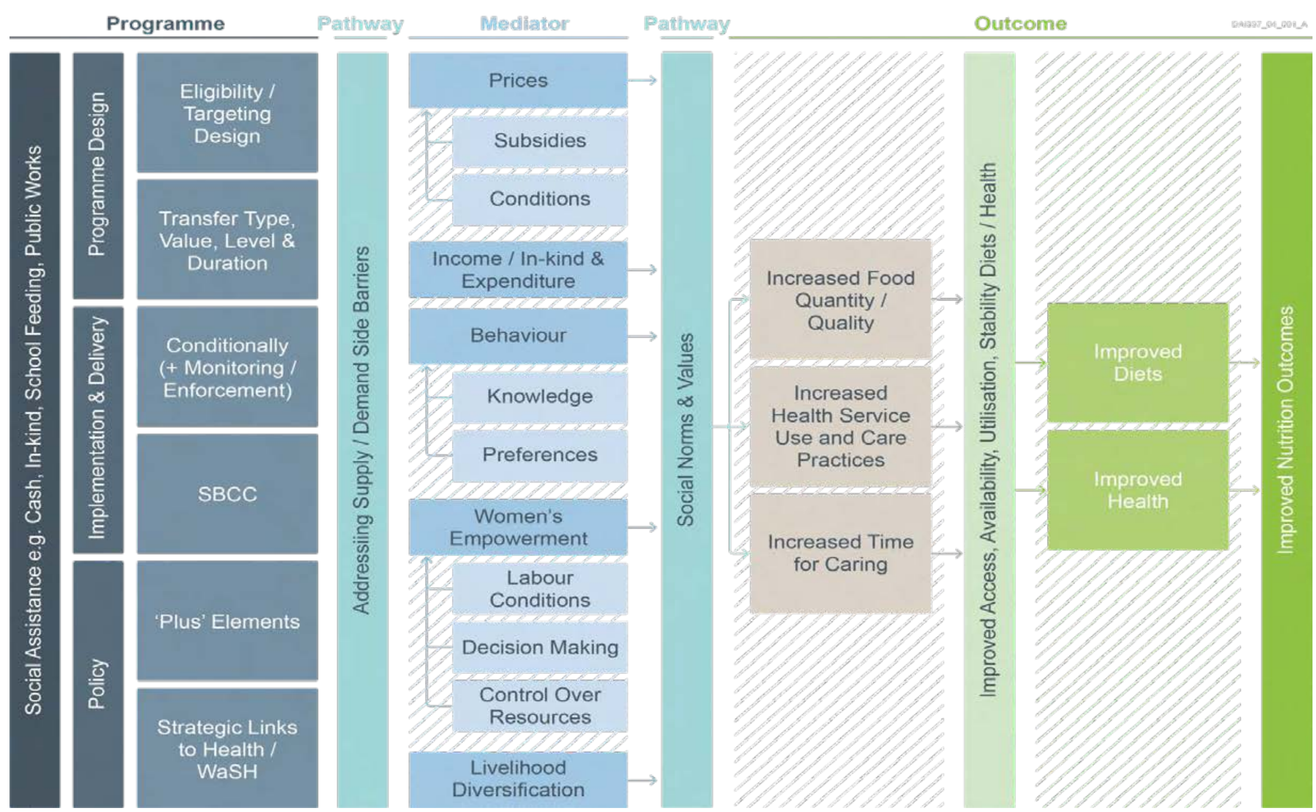
Social assistance programmes in the form of food assistance are intrinsically nutrition sensitive targeting food insecure and nutritionally vulnerable households. Whilst typically addressing the underlying determinants of malnutrition they may also address immediate determinates when combined with e.g., nutrition social and behaviour change (SBC). This section is particularly concerned with household food assistance in the form of in-kind food and CVA as platforms to support the nutrition status of PBWG.

In-kind food assistance e.g., General Food Distribution (GFD), is generally given to vulnerable households affected by crisis. GFD is provided as a food ration or basket (and may include fortified staples) which may be given to the whole community (blanket GFD) or targeted to those most vulnerable, depending on the situation and context.

Cash and Voucher Assistance (CVA) is the general term for programmes that provide either cash (unconditional (UCT) or conditional (CCT)), or food vouchers (FV) exchangeable for goods and services. They are provided directly to household recipients and may be part of a wider social protection programme.

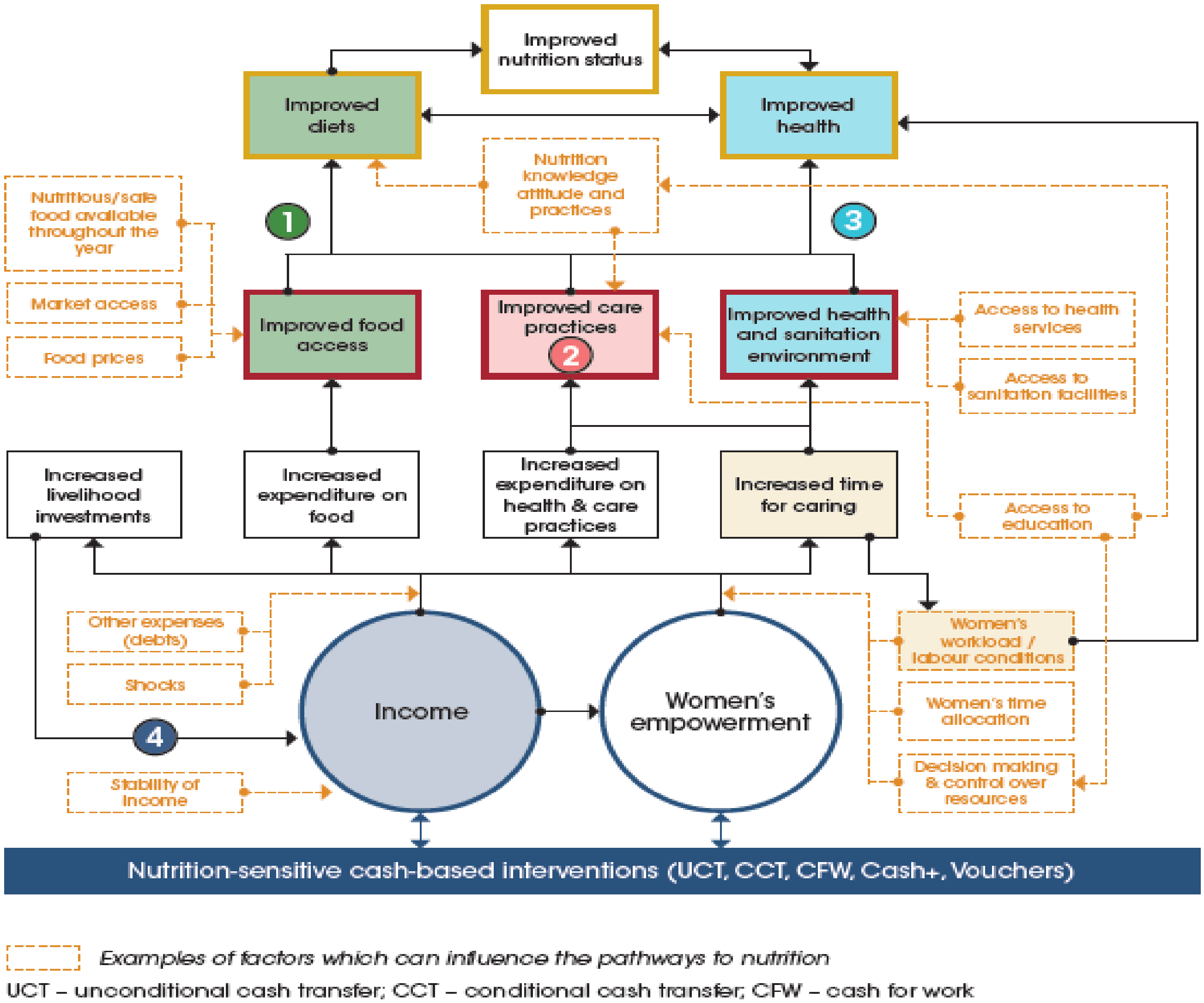
The pathways through which food assistance impacts nutrition outcomes has been captured in Figure 5 below.[122,123] Food assistance programmes can lead to improvements in household food security, access to health services and time available for care, ultimately leading to improved diets and health. However, the translation of these improvements into nutrition outcomes is not obvious, especially for PBWG. It is not entirely certain which of these pathways is more important because the evidence available is not sufficient to capture all these pathways in detail. For example, there is insufficient evidence around the different mediators identified, which themselves are not the objectives of food assistance, but which can be important barriers to improving nutrition outcomes.

FIGURE 5. **Indicative pathways from social assistance to nutrition outcomes**



Source: TASC. (2021). *Maximising nutrition outcomes through social protection policy and programming;* Alderman, H. (2016). Leveraging Social Protection Programs for Improved Nutrition: Summary of Evidence Prepared for the Global Forum on Nutrition-Sensitive Social Protection Programs. World Bank, Washington, DC.

FIGURE 6. **Potential impact pathways from nutrition-sensitive cash-based interventions to nutrition**



Source: Food and Agriculture Organization (2020). Nutrition and cash-based interventions – Technical guidance to improve nutrition through cash-based interventions adapted from *Bailey and Hedlund (2012), Herforth and Harris (2014) and SPRING (2014)*

A recent review of reviews and evidence suggests that cash and voucher assistance (CVA) programmes impact nutrition through two specific and complementary pathways i) as a source of income and ii) through women's empowerment (see Figure 6).[124]

However, although the evidence base for food-based SAPs is increasing in humanitarian settings[125] there still remain few robust studies, including on PBWG nutrition outcomes, from humanitarian settings[126] again suggesting that the evidence applied from pathways in development settings cannot be assumed to be the same in humanitarian settings.

What the evidence says

The evidence of the impact of SAPs in the form of food assistance on PBWG nutrition outcomes is scant, with more robust evidence on household dietary intakes compared to PBWG's own dietary intake.[127] Relating to PBWG and/or maternal nutrition outcomes only three recent reviews were identified.[128,129,130] These reviews captured most of the recent individual studies that have been carried out on food assistance and nutrition. But where deemed necessary i.e., where evidence was not summarised sufficiently, then individual papers were examined in more detail. As with agricultural-asset transfers, most of the available evidence comes from secondary objectives, with the primary objective on child nutrition outcomes. However, there was more specific reference to PBWGs.

Dietary intakes and diversity

Both in-kind and CVA interventions were seen to lead to improvements in PBWG and maternal dietary outcomes, diet quality and dietary diversity.[131,132,133] There have been a limited number of studies that have included measurements of micronutrient status and anaemia. However, there was some evidence of impacts on increasing haemoglobin concentration (Hb) and reduction in the prevalence of anaemia with *fortified* in-kind transfers, but not with cash.[134] However, there is also evidence of a negative impact of CVA (both cash and vouchers) on maternal Hb status and may have been due to the transfers being 'unconditional' and given without the necessary knowledge on diets and dependent on what was available in the markets.[135] Olney et al recommend that to reduce PBWG anaemia and increase Hb then transfers need to be fortified or supplemented, targeted to PBWG and possibly with an added SBC component.[136] As well they recommend including both household and individual transfers[137] so to ensure household food insecurity is not a barrier to PBWG's dietary intakes.

Anthropometry

Very few studies assessed programme impact on PBWG nutritional status outcomes in terms of BMI or MUAC. And the evidence that is available is mixed depending on the type of programme. For example, there is some evidence that has shown an improvement in maternal BMI, and a small improvement in maternal MUAC, with *fortified* in-kind transfers compared to CVA[138] and a positive impact of *fortified* in-kind transfers on MUAC during pregnancy, but not sustained postpartum.[139] Whereas other evidence on unfortified in-kind transfers saw no impact on maternal BMI.[140] Another review only pointed to an impact of CVA on maternal BMI if a nutrition SBC component was added.[141] An evaluation of Ethiopia's PSNP showed no impact on maternal BMI.[142] There is evidence that food vouchers, compared to cash, had a positive impact on maternal BMI but no impact on maternal MUAC.[143] On the flip side there is some evidence of a potentially negative impact on maternal BMI in contexts where the prevalence of overweight and obesity is high.[144] Overall, there is consensus that evidence around adolescent nutrition status is lacking.[145] Olney et al recommend that to improve BMI status the transfers need to be specifically targeted to women and with an added SBC component.[146] As well they recommend including both household and individual transfers[147] so to ensure household food insecurity is not a barrier to dietary intakes.

Women's empowerment

In general, it has been noted that women's empowerment and gender-based violence are the least explored outcomes of SAP in humanitarian settings. The limited evidence there is suggests that CVAs can have a positive impact on women's empowerment.[148] SAPs may also contribute to PBWG increased uptake of and access to antenatal care, health-seeking and nutrition services and nutrition information.

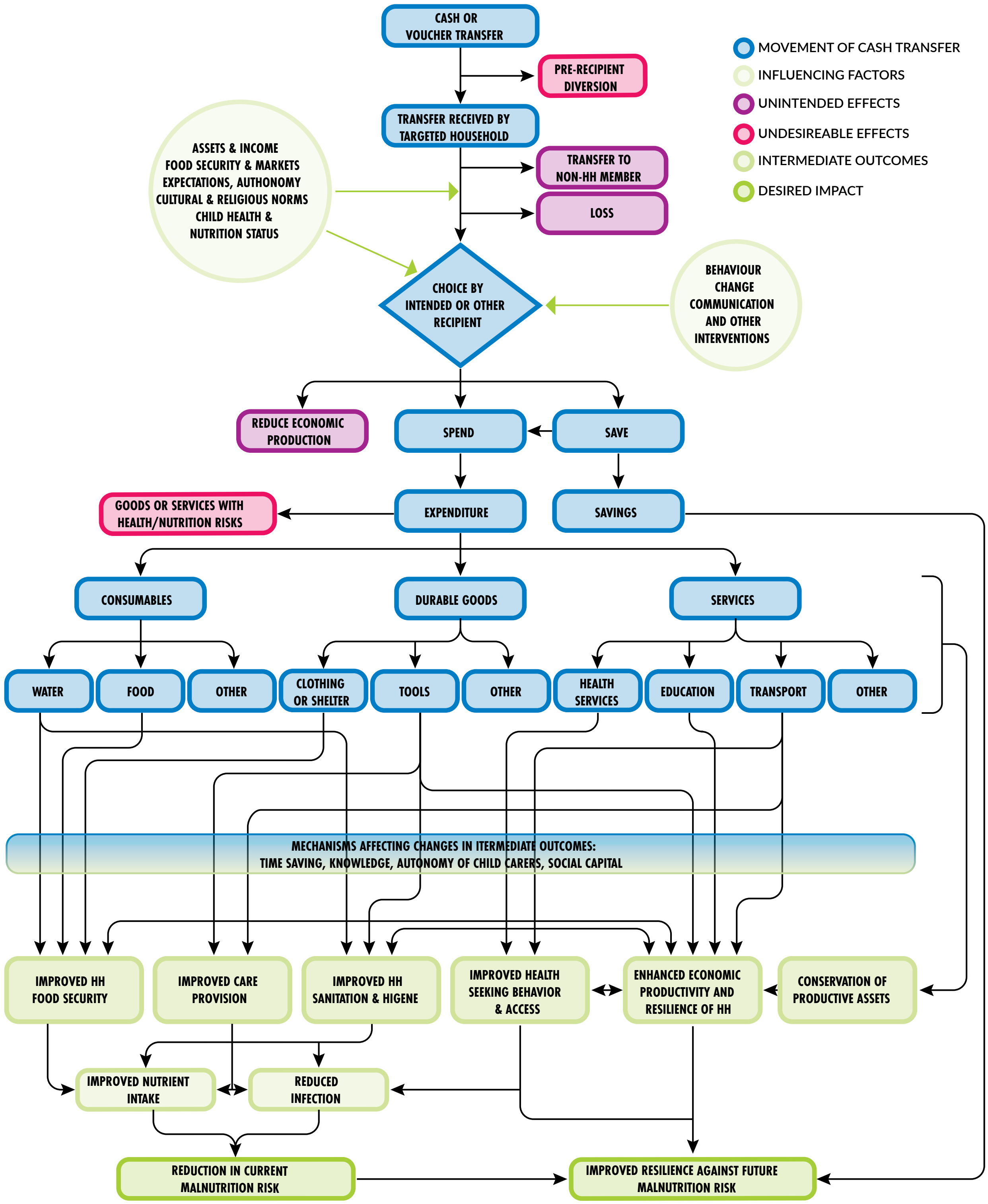
Research gaps

A research prioritisation exercise carried out in 2016/7 with the objective "to develop a research agenda on CTP [Cash Transfer Programming] for health and nutrition in humanitarian settings" prioritised 44 questions in the 'health and nutrition' category, of which, none specifically addressed women's nutrition (except for older women: *"What is the impact of multi-purpose, unconditional cash on older women and men's nutritional outcomes?"*)[149]. For PBWG there was one question; "*Does support to costs linked with health (e.g., transport, per diem for the caretaker, meals) contribute to better case management in particular for children and pregnant and lactating women?*". Searching on the term 'maternal' there were 2 questions – but only one related to nutrition: "*How do various types of cash transfers affect nutrition, HIV, and maternal health?*".

The same prioritisation exercise also carried out an evidence review which is available in an excel format. They included 48 reviews, case-studies and evaluations reported between 2015 and 2017; of which 19 had any nutrition focus. However, only 3 of those reported on women's nutrition,[150,151,152] with only the Fenn review citing pregnant or maternal.

Overall, there is a consensus that appropriate programme design, targeting and implementation is crucial for the success of a food assistance programme. If policy and programming are to be well informed then evidence is extremely important, especially in deciding which modality (e.g., type, value, level and duration) of food assistance e.g., CVA vs in-kind food transfers, or CVA only vs. CVA+. However, knowing the best way to implement a nutrition-sensitive SAP in any given context is difficult when faced with mixed and inconclusive context-specific evidence. In fact, it is the heterogeneity between studies, including modality, programme design features, implementation fidelity, objectives, that makes syntheses and meta-analyses of the data on food assistance programmes and nutrition outcomes difficult. One thing the above reviews do share in common is the call for more evidence generation with better designed evaluations and consistency in programme and evaluation designs across different programmes and contexts.

FIGURE 7. **REFANI theory of change**[153]



Source: Seal, A. Dolan, C. and Trenouth, L. (2017). Refani Synthesis Report. Research on Food Assistance for Nutritional Impact.

5. Women's empowerment/ resilience

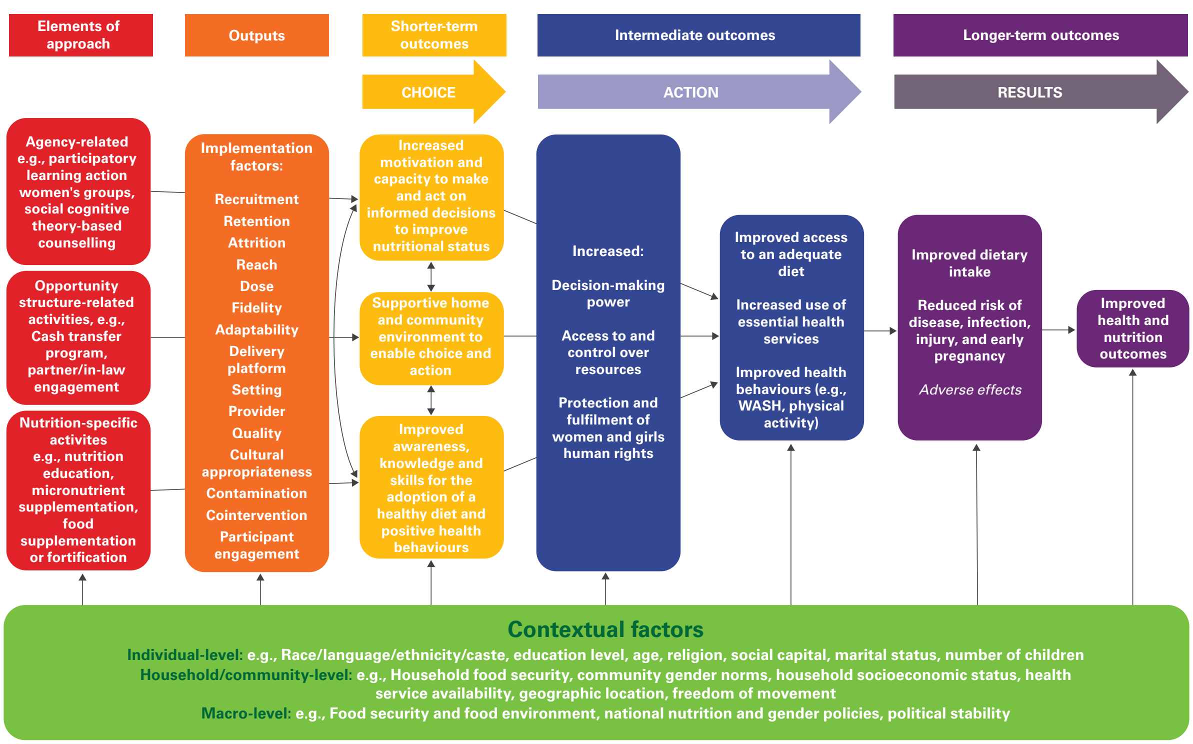
Introduction and impact pathway:

One definition of empowerment is *"the expansion in people's ability to make strategic life choices in a context where this ability was previously denied to them"*.[154]

The model below (from Riddle, 2021[155]) illustrates how women's empowerment approaches can result in improved nutrition outcomes. The 'elements of approach' column describes the necessary ingredients required to achieve these outcomes. Aside from the necessary nutrition- specific programmes available women also need 'agency' and 'opportunity' to achieve nutrition outcomes. If either is absent, then achievements will be sup-optimal. Nutrition interventions need to be designed to foster agency and support opportunities for maximum benefit. The 'contextual factors' include the critical cultural, policy and legal environment that restricts women's empowerment in many contexts and often needs to be addressed simultaneously.

There are a number of tools (Table 1) that have been developed to measure women's empowerment, agency and inclusion to understand where the barriers are in order to address them. Whilst these tools were developed primarily from the agricultural sector, nutrition modules, as well as market assessment modules are now included. A new emerging tool, called the abridged Women's Empowerment in Nutrition Index (A-WENI) has been piloted in rural South Asia and focuses more specifically on the aspects of empowerment that likely have significant bearings on nutrition outcomes. These tools have been designed to be lean and rapid and could be applied to the humanitarian context.

FIGURE 8. **Logic model: Empowerment-based nutrition interventions to improve women's nutritional status**



Source: Riddle, A. *et al.* (2021). PROTOCOL: The effects of empowerment-based nutrition interventions on the nutritional status of women of reproductive age in low- and middle-income countries. *Campbell Systematic Review,* 17(3).

TABLE 1. **Tool to measure women's empowerment, agency and inclusion in agriculture and nutrition programmes.**

Tool/Index		
WEAI	The Women's Empowerment in Agriculture Index	An aggregate index which reports empowerment at the country or regional level - based on individual-level data collected by interviewing women and men within the same households.
A-WEAI	Abbreviated WEAI	Building on the WEAI, the A-WEAI shortens the interview length and modifies questions that were difficult to implement in the field, while maintaining cross-cultural applicability.
Pro-WEAI	Project WEAI	Measures women's empowerment in various types of agricultural development projects and includes a module specific to nutrition and health projects. Includes both quantitative and qualitative tools to help projects understand local definitions of empowerment.
pro-WEAI+mi	Project-level WEAI for market inclusion	Measures empowerment across the value chain using both quantitative and qualitative tools.
WELI	Women's Empowerment in Livestock Index	Based on the WEAI framework the WELI facilitates assessments of the impact of livestock interventions on the empowerment of women involved in the sector.
WEN-Grid	Women's Empowerment in Nutrition	WEN Grid is a theoretically grounded tool for understanding empowerment and nutritional outcomes, combining Kabeer's empowerment framework and the UNICEF conceptual framework for causes of malnutrition.
WENI-Index	Women's Empowerment in Nutrition-Index	A nutrition-centred metric of empowerment that can be used to measure, track, and identify barriers to nutritional empowerment.
A-WENI	Abbreviated WENI	Can be incorporated in any general-purpose survey conducted in rural contexts.
WEFI	Women's Empowerment in Fisheries and Aquaculture Index	Inspired by the WEAI: measures the empowerment, agency, and inclusion of women in fisheries and aquaculture contexts to identify ways to overcome those obstacles and constraints. Uses both quantitative and qualitative tools.

More recently there has been greater acknowledgement of gender transformative approaches in humanitarian settings. Whilst there are valid reasons for projects to target by sex or gender (e.g., life-course effects of maternal malnutrition), there is a growing recognition in humanitarian and development spaces that gender does not mean only 'women and girls' but includes roles of and outcomes for men and boys.[156]

The Gender-Transformative Framework for Nutrition (GTFN) '*encourages humanitarian response and nexus programming that strengthens women, girls' and vulnerable communities' resiliency-building practices to respond to shocks. Women and girls must also be equipped with the knowledge, resources, and power to realize their rights in the midst of external shocks*'. The GTFN encompasses inner and outer rings (see Figure 9).[157] The outer ring breaks down the multi-sectoral dimensions of malnutrition while the inner rings seek to put women and girls at the centre of the process. The central placement of the empowerment circles enables users to understand the systemic impacts of gender inequality within each of these domains.

Whilst the importance of GTAs for nutrition has been acknowledged, most of the evidence, including operalisation of GTAs and frameworks (e.g., by nutrition practitioners and donors) comes from the agriculture sector rather than nutrition. What is available has been inconsistent and poorly documented, so making it difficult to understand 'what works'.

Summary of the evidence

- Generally a lack of consistent evidence
- Empowerment may act as a crucial driver of nutritional improvement.
- Agency and opportunity are essential ingredients to achieve nutrition outcomes
- There is a lack of clear understanding of the mechanisms through which empowerment affects household dietary diversity and nutrition.

FIGURE 9. **Domains of the Gender-Transformative Framework for Nutrition (GTFN)**

Source: O'Leary, M., Ameur, A.B., Anderson, S., Holte-McKenzie, M., Papastavrou, S. and Tse, C., 2020. A gender-transformative framework for nutrition. World Vision Canada.

What the evidence says

Reviews assessing intervention effects of nutrition programmes that adopt empowerment strategies are sparse especially so in humanitarian settings. However, for those studies that do exist there is a clear consensus of an association between women's empowerment and household, women and child nutritional improvements in different contexts.[158] At the same time, there is a lack of clear understanding of the mechanisms through which empowerment affects dietary diversity and nutrition.[159]

"Qualitative studies in Bangladesh and Nepal have found that women overwhelmingly cite discriminatory food allocation and low decision-making power as barriers to dietary intake during pregnancy".[160,161,162]

Working to decrease gendered barriers to school attendance (e.g., through menstrual hygiene management, latrines, prevention of early marriage and delaying first pregnancy) are important actions to prevent adolescent pregnancy and related nutrition challenges.

While implementing empowerment programmes in humanitarian settings may not always be feasible, gender should always be an integrated element in nutrition programming for PBWG and any humanitarian action should consider how it may inadvertently affect gender relations and power balances.

5.1 APPLYING MULTIPLE APPROACHES

Introduction and pathway

Packages of support are defined as more than one intervention provided to PBWG simultaneously.

WHO has placed substantial emphasis on nutrition assessment and provision of a set of nutrition interventions including provision of BEP supplementation, IFA and calcium supplementation, deworming, GWG monitoring and counselling on nutrition, family planning, and breastfeeding, coupled with efforts to prevent and treat maternal infections and anaemia.

Likewise, UNICEF's Maternal Nutrition Programming Guidance states UNICEF's commitment in emergencies to support a minimum package of preventive interventions for all PBWG, including fortified staple foods, counselling on nutritious diets, MMS or IFA supplements, and deworming prophylaxis.

In settings where there is food insecurity and poor access to diverse and quality diets, UNICEF will also support services to prevent maternal wasting as part of national systems. In these situations, UNICEF will provide targeted preventive nutritional supplements (i.e., BEP) to PBWG who have been screened and referred. UNICEF will also create opportunities to provide food vouchers and cash-based interventions for women and link these with nutrition counselling.

What the evidence says

Summary:

There is evidence of some promising practices of delivering packages of interventions or integrated interventions to improve PBWG nutrition, though as yet a robust body of evidence across a variety of contexts is lacking.

A post-intervention quasi-experimental study was conducted[163] among pregnant women in Rwanda to determine the effect of an integrated intervention on their nutritional status in highly food insecure districts. It was implemented in two intervention and two control districts. The integrated nutrition intervention package comprised one component of nutrition-specific intervention, which was NEC and three nutrition-sensitive components: promotion of agricultural productivity, promotion of financial literacy and economic resilience and improved access to WASH services. Maternal undernutrition was defined as low MUAC (<23 cm) during delivery or low BMI (<18.5 kg/m2) in the first trimester or both. After controlling the potential confounders using multivariable logistic regression, maternal undernutrition was 77% times less likely to occur among pregnant women in the intervention group compared with those in the control group (adjusted odds ratio [aOR] = 0.23; 95% confidence interval [CI] = 0.15–0.36; $p < 0.001$). The authors concluded that integrated nutrition-sensitive and nutrition-specific interventions can reduce maternal malnutrition, such as undernutrition and anaemia.

A literature and programming review[164] on how to improve the nutrition status of adolescent girls in Pakistan, concluded that health workers may not be best placed to reach adolescent girls. The study found poor uptake of services by adolescent girls, with important barriers including negative judgement by service providers, resistance to consult health service providers by the girls' families, especially their fathers, and evidence suggesting that a quarter of married adolescents avoided health facility ANC provision, citing preference for a female doctor. The authors proposed that insights are needed to better understand suitability of media channels for the communication of potentially sensitive messages across different contexts, citing a study across three districts that found that although a high proportion of participants watched TV, girls preferred parents, community health workers and teachers as their source of information on sexual and reproductive health issues.

The review found that different approaches, delivery platforms, packaging, and amount of nutrition messages targeting adolescent girls are needed from those targeting older women; additionally noting that adolescent girls are not a homogenous group and that large differences exist across contexts, such as in urban versus rural areas, across different wealth quintiles, and different geographic settings. The study concluded that to ensure social change, and influence social norms around adolescent girls, interventions must also target adolescent boys, as well as parents and the community at large. The authors' recommendations to scale up action to improve nutrition of adolescent girls include: filling the knowledge gap, through conducting surveys on adolescent nutrition, and investing in formative research addressing adolescent food and nutrition behaviours and determinants; bringing together multi-sectoral alliances of organisations and services that work with adolescents, to integrate and strengthen targeted nutrition interventions into economic livelihood development, education, and gender and life skills development programmes; testing potential nutrition interventions for adolescents through existing cross-sectoral platforms, as well as developing new platforms, such as community clubs or social media.

A systematic review[165] searched studies on women and children receiving nutrition-specific interventions during or within five years of a conflict in LMICs. General food distribution (n=34), micronutrient supplementation (n=27) and nutrition assessment (n=26) were the most frequently reported interventions, with most reporting on intervention delivery to refugee populations in camp settings (n=63) and using community-based approaches. Only eight studies reported on coverage and effectiveness of interventions. Most studies examined programmes for children and there was very little reported on programming for women. The review concluded that there is very little information on coverage or effectiveness of nutrition interventions; more rigorous evaluation of effectiveness and delivery approaches is needed, including outside of camps and for preventive as well as curative nutrition interventions.

A secondary analysis of data provided through a larger birth cohort study conducted in rural Ethiopia (3 woredas (districts) of Oromiya) between 2014 and 2016 was conducted.[166] Pregnant women (n = 4680) were recruited using a rolling enrolment and surveillance approach. A significant association was found between low MUAC and the household WASH score (P = 0.017): for every unit increase in the household WASH score, the odds of a low MUAC decreased. Compared to a household with a poor WASH score, households in the fair WASH category had 20% reduced odds of a low MUAC. Both the MDD and MUAC in women improved as the household WASH scores improved. The study concluded that interventions aimed at enhancing the diet and nutritional status of women during and after pregnancy should include relevant WASH components as essential elements in multisector nutrition programming.

A desk review by Sethi (2021) of studies in South Asia on management of thinness in pregnancy reported that only two countries have national guidelines describing packages of support for pregnant women: Sri Lanka and Nepal[167,168,169]. In Sri Lanka, assessment of maternal nutritional status using pre-pregnancy BMI is mandatory. Any pregnant women with BMI <18.5 or inadequate weight gain (<1 kg/month) is enrolled for a special service package that includes assessment of calorie, protein, and micronutrient intake using 24-h dietary recall; assessment of dietary habits, food taboos, misconceptions; nutrition counselling on increasing the amount of starch-based foods at each meal, consuming 1–2 extra meals than usual, using 1–2 tablespoons of oil and including fish/dried fish/egg, pulses, vegetables, and green leaves to the daily diet; dietary supplementation with Thriposha *(50 g/day to be consumed with two teaspoons each of sugar and coconut: 401.8 kcal, 61.9 g carbohydrate, 20 g protein, 18 mg iron);* and follow-ups to assess compliance and, as for other pregnant women, prevent anaemia including IFA supplementation and deworming.[170] Thin pregnant women and their husbands/other family members are counselled on non-diet related themes, including reducing workload, mental relaxation, foetal well-being, limiting exposure to household smoke and passive tobacco smoke, and examination conducted for any other illnesses in mothers such as urinary tract infections, parasitic infections, and medical illnesses (hypertension, diabetes). If weight gain continued to be inadequate, pregnant women were referred to a higher health facility.

In Nepal's Integrated Management of Acute Malnutrition (IMAM) guidelines, pregnant women (second and third trimester) and breastfeeding women (children under 6 months) with MUAC<23 cm are nutritionally managed by; (a) Nutrition counselling on use of locally available foods, preparation and intake of high energy-dense meals—such as *Sarbottam Pitho*, a locally developed "super flour"—to provide extra daily nutrient requirements, provision of MMNs to fortify foods at home; (b) In areas where local food availability and access are not sufficient to improve nutritional status, provision of 200 g of dry FBF per day made from a mixture of corn, wheat, rice, soya, milk powder, sugar, oil, vitamins, and minerals (787 kcal, 33 g protein [17%], 20 g fat, essential fatty acids and all the required micronutrients), along with counselling on use of supplementary food.

Research gaps

There are some promising examples of integrated or combined packages of support for improving nutrition outcomes for PBWG. However, rigorously designed trials and evaluations of programmes are required to build the evidence required to develop guidance for PBWG nutrition in humanitarian settings.

Research gaps and limitations

A few themes recur throughout all the interventions examined to tackle PBWG nutrition:

- Too few studies have examined PBWG outcomes – the focus has predominantly been on child outcomes; or birth outcomes for pregnancy-specific interventions.
- Very few studies examined programme impacts on nutrition status of PBWG using MUAC (or BMI); a few examined anaemia.
- The majority of available evidence on nutrition interventions for PBWG comes from stable contexts and there is a lack of evidence of what works in humanitarian contexts, especially in conflict settings.
- Where evidence exists, the heterogeneity between studies, including modality, programme design features, implementation fidelity, objectives, makes syntheses and meta-analyses of the data on intervention programmes and nutrition outcomes difficult.
- Very few studies have differentiated pregnant adolescents from pregnant adult women in programme design or in measuring outcomes.
- There is a dearth of studies examining nutritional status or outcomes of interventions for breastfeeding women.
- Evidence across the majority of interventions points to the need to consider individual approaches in combination with household/family approaches and/or community approaches to improve effectiveness on PBWG nutritional outcomes.

Recommendations for research priorities

There is insufficient information to determine whether MUAC can be used as an indicator to predict the potential benefit (e.g., improved foetal growth, improved outcomes for PBWG) **of a certain nutritional intervention.** More MUAC data needs to be collected over the course of implementation of interventions and at outcome to establish this role.

BEP: future trials should focus on PBWG at risk of or underweight, provide BEP intervention that follows an agreed protocol (in terms of ration size, composition, delivery modality and duration), compare BEP with MMN or LNS, and ideally employ an RCT design. BEP and LNS trials should report the dietary intake and total calorie intake of study arms to clarify whether BEP and LNS are replacing or increasing the calories in the diet.

Studies should also examine what is the best composition of BEP (not restricted to products but including local foods) for averting which risks.

More implementation research on engaging family members and communities in PBWG nutrition is needed[171].

Research is urgently needed that collects longitudinal data on both women and adolescents from pre-conception through pregnancy and breastfeeding to better understand and inform differentiated MUAC cut-offs for breastfeeding women and for pregnant and breastfeeding adolescent girls.

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